QFI and method of moments: $N = 60$, $B/J = 0.40$

QFI, F_h — $\mathcal{M}_h[\hat{S}_z]$ — $\mathcal{M}_h[\hat{X}]$ — $\mathcal{M}_h[\hat{Y}]$, $\hat{Y} = i\hat{S}_z\hat{X}$

(a) $\left(\left|E_1\right\rangle + \left|E_{\overline{1}}\right\rangle\right)/\sqrt{2}$

(b) $\left|E_1\right\rangle$

$\frac{2\pi}{\Delta_{7\overline{7}}}$ $\frac{2\pi}{\Delta_{6\overline{6}}}$ $\frac{2\pi}{\Delta_{5\overline{5}}}$ $\frac{2\pi}{\Delta_{4\overline{4}}}$ $\frac{2\pi}{\Delta_{3\overline{3}}}$ $\frac{2\pi}{\Delta_{2\overline{2}}}$ $\frac{2\pi}{\Delta_{1\overline{1}}}$

$\mathcal{M}_h/(Nt)^2$

0.32
0.24
0.16
0.08
0.00

(c) $\left|\uparrow\cdots\uparrow\right\rangle$

(d) $\left(\left|\uparrow\cdots\uparrow\right\rangle + \left|\downarrow\cdots\downarrow\right\rangle\right)/\sqrt{2}$

$F_h/(Nt)^2$,

0.32
0.24
0.16
0.08
0.00

10^0 10^3 10^6 10^9 10^{12} 10^{15} 10^{18}

t/T

$\mathcal{M}_h[\hat{O}] = (\partial_h\langle\hat{O}\rangle)^2/\mathrm{Var}(\hat{O})$